

Himanshu Kurapati
Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend Software Engineer with 4+ years of experience specializing in **cloud-native migration** and **event-driven microservices development** across automotive and fintech domains. At Ford, led the migration of legacy on-premise vehicle telemetry pipelines to **AWS EKS** and **Kafka**, improving data throughput by 30%. At Capgemini, owned end-to-end delivery of a **loan origination module**—from legacy SQL to MongoDB migration to OAuth 2.0 secure API layer—processing 50K+ daily transactions. Proven ability to take full module ownership, optimize database performance, and deploy secure, real-time backend systems at scale.

TECHNICAL SKILLS

- **Languages:** Java, Python, TypeScript, SQL, Go
- **Frameworks:** Spring Boot, Django, FastAPI, Node.js
- **Cloud:** AWS (EKS, EC2, S3, Lambda, RDS, DynamoDB, MSK, CloudWatch), Azure (App Service, Key Vault, Functions), GCP (Cloud Run)
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Amazon RDS
- **Messaging & Streaming:** Apache Kafka, AWS MSK, RabbitMQ
- **DevOps:** Docker, Kubernetes (EKS), Helm, Terraform (basic), Jenkins, GitHub Actions, Azure DevOps, ArgoCD (basic)
- **AI/ML Integration:** OpenAI API, Bedrock (basic), LlamaIndex, embeddings for semantic search, fraud scoring with XGBoost (basic)
- **Testing:** JUnit, Testcontainers, Postman, Locust, JMeter, SonarQube
- **Monitoring:** Prometheus, Grafana, OpenTelemetry (basic)
- **Tools:** Git, Jira, Confluence, Figma (API specs)

CERTIFICATIONS

- **Microsoft Certified:** Azure Fundamentals (AZ-900)
- **AWS Certified:** Cloud Practitioner
- **Certified Kubernetes Application Developer (CKAD)**

WORK EXPERIENCE

Software Engineer

Ford Motor Company | Dearborn, MI | Mar 2024 – Present

Domain: Connected Vehicle Telemetry & Cloud Migration

- Led end-to-end migration of legacy vehicle telemetry ingestion pipeline from on-premise servers to **AWS EKS**, reducing infrastructure costs by 25% and improving data availability to 99.95%.
- Designed and deployed **12+ Java Spring Boot microservices** handling real-time vehicle diagnostics, increasing backend event processing throughput from 8K to 11K events/second (~30% improvement).
- Implemented Kafka producers and consumers using **AWS MSK** to process 2M+ daily telemetry events, enabling real-time anomaly detection for connected vehicle commands.
- Built secure REST APIs with **OAuth 2.0 and JWT** integrated with AWS Cognito, reducing unauthorized access attempts by 95% across internal and partner integrations.
- Containerized all backend services using **Docker** and deployed to **EKS with Helm charts**, achieving zero-downtime rolling updates and reducing deployment failures by 40%.
- Optimized DynamoDB table design and Redis caching for vehicle command workflows, reducing average API response latency from 210ms to 145ms (31% improvement).

- Created CI/CD pipelines using **GitHub Actions and Jenkins** with automated canary deployments, cutting release lead time from 2 days to 4 hours.
- Integrated **OpenTelemetry** with CloudWatch and Grafana for distributed tracing, reducing mean time to detection (MTTD) for production issues by 50%.
- Collaborated with frontend teams to expose **WebSocket endpoints** for React-based real-time dashboards, supporting 500+ concurrent dealer connections.

Environment: Java 17, Spring Boot 3, AWS (EKS, MSK, DynamoDB, Cognito, CloudWatch), Kafka, Redis, Docker, Helm, Jenkins, GitHub Actions, OAuth 2.0, JWT, OpenTelemetry, Grafana

Software Engineer

Capgemini | Hyderabad, India | Feb 2022 – Dec 2023

Domain: FinTech Loan Processing & Application Modernization

- Owned end-to-end delivery of **loan origination module** handling 50K+ daily transactions, from requirements to production deployment for a top Indian banking client.
- Migrated 2M+ customer records from legacy **SQL Server** to **MongoDB**, reducing query latency for credit check APIs by 40% and improving write scalability during peak hours.
- Designed and implemented **15+ REST APIs** for KYC verification, credit scoring, and payment gateway integrations using **Django and Java Spring Boot**, achieving 99.5% uptime.
- Integrated **Azure Key Vault** for secrets management and implemented field-level encryption for PII (PAN, Aadhaar), passing internal security audit with zero findings.
- Built event-driven async processing for loan approval workflows using **Azure Service Bus** (Kafka-equivalent), reducing average workflow completion time from 3 minutes to 45 seconds.
- Developed **fraud scoring microservice** using a lightweight XGBoost model (trained offline) to flag suspicious applications in real time, reducing false positives by 20%.
- Refactored monolithic payment validation logic into **3 reusable microservices**, improving deployment frequency from monthly to bi-weekly and reducing merge conflicts by 60%.
- Performed load testing with **Locust** (10K concurrent users) and optimized database indexes, increasing peak transaction handling from 200 to 350 requests/second.
- Collaborated with QA and DevOps in **Agile/Scrum** to write 85%+ unit and integration test coverage using JUnit and pytest, reducing production bugs by 30%.

Environment: Python, Django, Java 11, Spring Boot, MongoDB, SQL Server, Azure (App Service, Key Vault, Service Bus), REST APIs, XGBoost (basic), Locust, JUnit, Azure DevOps, Git, Jira

EDUCATION

- **Master of Science in Computer Science** | Northern Arizona University, Flagstaff, AZ | Jan 2024 – May 2025
- **Bachelor of Technology in Computer Science** | Marri Laxman Reddy Institute of Technology and Management, Hyderabad, India | Aug 2018 – Sep 2022